

PLSC 30600

Week 9: External validity (generalizability + transportability)

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Roadmap

- What is external validity? Study sample vs. target population.
- When do the study ATE and target ATE differ?
- Identification + methods: standardization and (inverse-odds) weighting.
(Cole and Stuart, 2010)
- Diagnostics: overlap, extreme weights, effective sample size.
- Why this is hard even with RCTs: mechanisms, support factors, and
“licenses to transport.” (Deaton and Cartwright, 2018)

Internal vs. external validity

- **Internal validity:** does D cause Y *in this study*?
- **External validity:** what can we say about effects *elsewhere*?
- Two reasons estimates change across settings:
 - **Effect heterogeneity:** $\tau(x)$ differs across people/contexts.
 - **Selection:** the study sample is not a random draw from the target.
 - **(SUTVA/Design differences:** the treatment or its implementation differs across settings.)

Deaton & Cartwright emphasize: an RCT answers a well-defined question for the trial sample; using it for policy/general claims is a *separate* inferential step. (Deaton and Cartwright, 2018)

A clean setup: study sample vs. target population

Let $S_i \in \{0, 1\}$ indicate where unit i comes from:

- $S_i = 1$: in the study (trial) sample.
- $S_i = 0$: in the target population sample.

Define two estimands:

$$\tau_S \equiv E[Y(1) - Y(0) \mid S = 1] \quad \text{and} \quad \tau_T \equiv E[Y(1) - Y(0) \mid S = 0].$$

If effects are constant, $\tau_S = \tau_T$. If effects vary and selection shifts the X -mix, they can differ a lot.

Why $\tau_S \neq \tau_T$: heterogeneous effects + different composition

Let X be an effect moderator. Then

$$\tau_S = \sum_x \tau(x) \Pr[X = x \mid S = 1], \quad \tau_T = \sum_x \tau(x) \Pr[X = x \mid S = 0].$$

Even with perfect internal validity in $S = 1$, a different X -distribution changes the average effect.

Stratum		$\tau(x)$	$\Pr[X = x \mid S = 1]$	$\Pr[X = x \mid S = 0]$
$X = 0$	0		0.8	0.2
$X = 1$	5		0.2	0.8
τ			1.0	4.0

Three distinct external validity questions

- **Generalizability:** from a study sample to the population it was meant to represent.
- **Transportability:** from one population/context to a different one.
- **Extrapolation:** beyond the observed support (new X , new treatment version, new setting).

Key idea: “external validity” is not a binary property of a design; it depends on the *target* and the *assumptions* you are willing to defend. (Deaton and Cartwright, 2018)

Identification assumptions for τ_T (potential outcomes)

Two pieces:

- **Internal validity within the study** (e.g., random assignment):

$$D \perp\!\!\!\perp (Y(1), Y(0)) \mid S = 1.$$

- **External validity / sampling ignorability:**

$$(Y(1), Y(0)) \perp\!\!\!\perp S \mid X,$$

plus **overlap** $0 < \Pr[S = 1 \mid X = x] < 1$ where you want to generalize.

Overlap is checkable; the key ignorability claim is not. The substantive burden is: have we measured the right moderators X ? (Cole and Stuart, 2010) (Tipton, 2013)

What's testable vs. what's not

- **Testable / diagnosable:**

- Differences in X across $S = 1$ vs. $S = 0$ (and overlap).
- Extremity of weights; effective sample size; balance after weighting.

- **Not testable from the study alone:**

- Whether you captured all effect moderators in X .
- Whether the “same treatment” has the same meaning/version in the target.
- Whether support factors/mechanisms differ across settings.

Cole & Stuart: sampling propensity scores and inverse-odds weights

If we observe X in both samples, we can model trial participation:

$$e_S(x) \equiv \Pr[S = 1 \mid X = x].$$

Then reweight the *study* sample to match the *target* X -distribution using inverse-odds weights:

$$w_i \propto \frac{\Pr[S = 0 \mid X_i]}{\Pr[S = 1 \mid X_i]} = \frac{1 - e_S(X_i)}{e_S(X_i)}.$$

- Intuition: upweight study units who look “target-like” on X .
- This is direct standardization in disguise (and closely related to survey weights).
- Inference: default to robust (sandwich) SEs for weighted estimators. (Cole and Stuart, 2010)

A simple estimator for τ_T (weighted RCT within $S = 1$)

Under sampling ignorability and overlap,

$$\tau_T = \mathbb{E} [\mathbb{E} [Y(1) - Y(0) \mid X, S = 1] \mid S = 0] .$$

One implementation (Cole-Stuart / Tipton):

- Estimate $e_S(X)$.
- Compute weights $w_i \propto (1 - \hat{e}_S(X_i)) / \hat{e}_S(X_i)$ for $S = 1$ units.
- Estimate τ_T as a weighted treated-control difference *inside the trial*:

$$\hat{\tau}_T = \bar{Y}_{D=1,w} - \bar{Y}_{D=0,w} .$$

Outcome-model standardization (and doubly robust variants) are alternative implementations of the same idea: average over the target X -distribution. (Cole and Stuart, 2010) (Tipton, 2013)

Precision intuition: extreme weights $\Rightarrow$ tiny effective sample size

With weighting, precision often depends more on the *effective sample size* than the raw n :

$$\text{ESS} \equiv \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

- When study and target are similar in X , weights are mild and ESS is close to n .
- When study and target are far apart, weights become extreme and ESS collapses.
- With no overlap, transport fails: you are extrapolating beyond the data.

Diagnostics checklist

- Define the target estimand first (what population/context?).
- Compare X distributions across $S = 1$ and $S = 0$.
- Plot the sampling propensity score and check overlap.
- Inspect weight distributions (max, tail behavior, ESS).
- Check balance after weighting (e.g., standardized mean differences).
- Sensitivity: different moderator sets X , functional forms, trimming/restriction.

Applied demo: NSW trial $\rightarrow$ CPS target

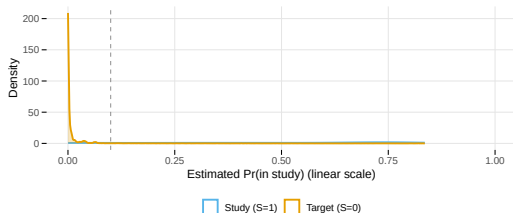
Demo setup

- Study sample $S = 1$: the NSW job-training experiment (nsw).
- Target sample $S = 0$: CPS covariate distribution (cps1).
- Goal: estimate the ATE for the target using inverse-odds weights.

This is the Cole–Stuart logic in a simple difference-in-means setting (no survival model): estimate $\Pr[S = 1 \mid X]$, weight the trial, then re-estimate the treatment effect. (Cole and Stuart, 2010)

Sampling propensity score overlap (linear scale)

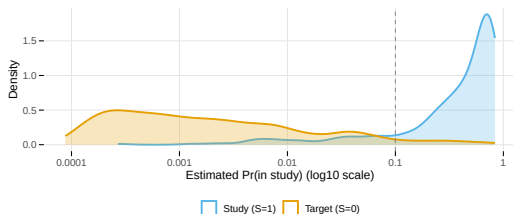
First, the standard plot on a 0–1 x-axis. On this scale, most of the target mass is piled near 0, so the overlap is hard to see.



Sampling propensity score overlap (log scale):

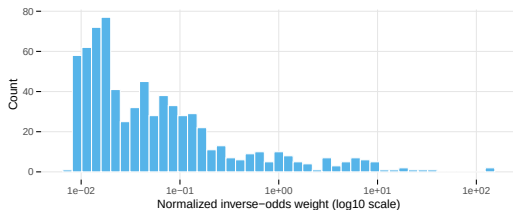
$$\hat{e}_S(X) = \Pr[S = 1 \mid X]$$

If overlap is weak, the weights needed to match the target become extreme.



Inverse-odds weights in the NSW study sample

Weights are $w_i \propto (1 - \hat{e}_S(X_i))/\hat{e}_S(X_i)$ for $S = 1$ units.



What happens here? (a cautionary example)

- NSW SATE (study ATE): $\hat{\tau}_S \approx 886$.
- Weighted estimate targeting CPS: $\hat{\tau}_T \approx -4627$.
- Effective sample size under weights: $ESS \approx 10.6$ (out of $n = 722$).

Interpretation: most of the CPS target population looks unlike the NSW study sample on observed covariates, so the generalization attempt relies on a handful of heavily-weighted study units.

Two honest responses to weak overlap

- **Redefine the target:** transport only to the overlap region (common support).
- **Or keep the target but accept uncertainty:** trim/regularize weights and show sensitivity.

Neither option “fixes” missing moderators or context changes; it just makes the statistical problem better behaved.

An overlap-restricted target (illustration)

Restrict the CPS target to units with $\hat{e}_5(X) \geq 0.1$, i.e., units that look plausibly trial-eligible on observed X :

- Size of restricted target: 802 ($\approx 5\%$ of CPS).
- Weighted estimate for this restricted target: $\hat{\tau}_{T,\text{overlap}} \approx 374$.
- ESS under overlap weights: ≈ 201.7 .

Message: statistical generalization is often only credible for a *narrower* target than we hoped.

Tipton: the two bridges (statistical + substantive)

- **Statistical bridge:** make $S = 1$ look like $S = 0$ on moderators X (weighting/stratification).
- **Substantive bridge:** argue that these X variables are the right moderators and that mechanisms are stable.

Tipton emphasizes **subclassification** on the sampling propensity score as a stable alternative to raw IPW.

(Tipton, 2013)

Multi-site experiments: heterogeneity is the point

- Think of sites as contexts: treatment effects may vary by site and site covariates.
- External validity goals:
 - learn $\tau(\text{site})$ or $\tau(x)$ (moderation), then average over a target mix;
 - quantify heterogeneity and uncertainty (partial pooling / hierarchical models).
- Warning: sites are rarely a random sample of “all sites” $\Rightarrow$ site selection is an external-validity problem too.

Design-based external validity (what to do before you estimate)

- Build a sampling frame for the target population.
- Use probability sampling (or purposive oversampling) of key subgroups/contexts.
- Block/stratify on plausible moderators to learn heterogeneity.
- Measure baseline covariates you will need for standardization later.
- Report the target explicitly in the paper (who/where/when).

Why transport is not automatic (even with RCTs)

Deaton & Cartwright: what RCTs do (and do not) buy you

- Randomization helps with internal validity, not automatic generalization.
- External validity depends on **invariance**: which causal relationships are stable across settings.
- “Same treatment” may not be the same intervention (versions, compliance, spillovers).
- Causes can require **support factors** (enabling conditions) that differ across contexts.
- Randomization balances covariates *in expectation*, not necessarily in a realized finite sample.

The reading's main message: external validity is not a checkbox; it is an argument about mechanisms + context. (Deaton and Cartwright, 2018)

Support factors (mechanisms) as “unmeasured moderators”

- Weighting/standardization assumes we measured the moderators X .
- Deaton–Cartwright emphasize that important moderators are often **contextual**:
 - institutions, implementation capacity, norms, market responses, networks/spillovers.
- If those differ across settings and are unmeasured, reweighting can still fail.

Good practice: say what you would need to observe (or what additional study you'd run) to justify transport.

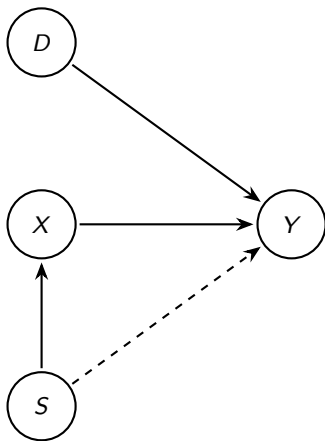
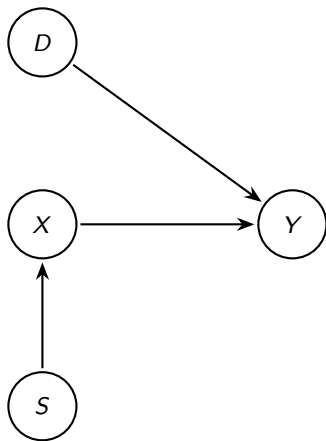
Pearl & Bareinboim: selection diagrams and “licenses to transport”

Pearl & Bareinboim: transportability as identification

- A selection diagram is a DAG plus nodes S indicating what differs across populations.
- Transportability asks: can we write the target causal effect using (i) study causal knowledge + (ii) target covariate information?
- The goal is explicit assumptions (“licenses to transport”), not a list of threats.

We'll keep this high level; the big idea is: sometimes standardization is sufficient, sometimes you need additional measurements, and sometimes it is impossible without new data. (Pearl and Bareinboim, 2014)

Selection diagrams: when reweighting is (and isn't) enough



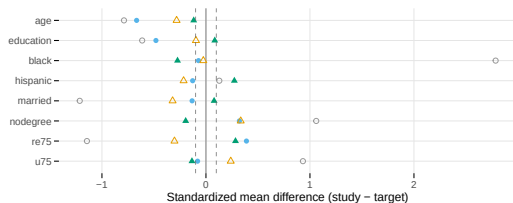
- Left: if populations differ only in the distribution of X , standardization/weighting can work.
- Right: if the outcome mechanism differs across populations (dashed $S \rightarrow Y$), reweighting on X is not sufficient.

Backup: code sketch for the demo (NSW → CPS)

```
> # NSW trial (S=1) -> CPS target covariates (S=0)
> Xvars <- c("age", "education", "black", "hispanic", "married", "nodegree", "re75", "u75")
> comb <- rbind(transform(nsw, S = 1)[, c("S", Xvars)], transform(cps1, S = 0)[, c("S", Xvars)])
> sel_fit <- glm(S ~ ., data = comb, family = binomial())
> ps <- predict(sel_fit, newdata = nsw, type = "response")
> w <- (1 - ps) / ps
> w <- w / mean(w) # normalize so mean weight = 1
> tau_tate <- with(nsw, weighted.mean(re78[treat == 1], w[treat == 1]) - weighted.mean(re78[treat == 0], w[treat == 0]))
> tau_tate
```

Backup: balance diagnostics (standardized mean differences)

- Compare study vs target on X : before and after weighting.
- With weak overlap, “good balance” can come from a few extreme weights $\Rightarrow$ always pair balance with ESS/weight diagnostics.



erlap CPS) ● Weighted (full CPS) ▲ Weighted (overlap CPS) ● Unweighted (full CPS) ● Unweighted (o

References I

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- Deaton, A. and Cartwright, N. (2018). Understanding and misunderstanding randomized controlled trials. *Social Science & Medicine*, 210:2–21.
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- Tipton, E. (2013). Improving generalizations from experiments using propensity score subclassification: Assumptions, properties, and contexts. *Journal of Educational and Behavioral Statistics*, 38(3):239–266.